



PES UNIVERSITY, BANGALORE

(Established under Karnataka Act 16 of 2013)

Department of Science and Humanities

B Tech: Second Semester

Session: Feb-May 2025

Subject Name: Engineering Mathematics II (4-0-0-4-4)

Subject Code: UE24MA141B

MATLAB - PRACTICE PROBLEMS

Unit 2

Laplace Transform

1. Solve the Initial Value Problem (IVP) using the Laplace Transform

$y'' + 5y' + 6y = 0$ using $y(0) = 1, y'(0) = 0$

output: $Y_s = (s + 5)/(s^2 + 5s + 6)$

$y_t = 3 \cdot \exp((-2 \cdot t)) - 2 \cdot \exp((-3 \cdot t))$

2. Solve the Boundary Value Problem (BVP) using the Laplace Transform

$y'' = -\pi^2 y, \quad y(0) = 0, y(1) = 0$

output: solution = 0

3. $f(t) = u(t-3) \cdot (t-3)^2$, where $u(t)$ is the unit step function solve using MATLAB code

output: $F_s = (2 \cdot \exp((-3 \cdot s)))/s^3$

4. Solve the inverse laplace function of $F(s) = s^2 + 2s + 1$ in MATLAB code

output: $\text{dirac}(t) + 2 \cdot \text{dirac}(1, t) + \text{dirac}(2, t)$